# AI-Generated Video and Audio: Analysing the Effectiveness of Generative AI in Creating Personalised Digital Content

## Abstract

The rapid advancement of generative artificial intelligence has fundamentally transformed the creation of personalised digital media. This literature review systematically evaluates the effectiveness of generative AI in producing customised video and audio content across three critical dimensions: content quality, production efficiency, and ethical challenges. Drawing from peer-reviewed sources published between 2020 and 2026, the review synthesises empirical evidence demonstrating that AI-personalised videos and audio content achieve significant engagement gains—with personalised audio ads increasing brand favourability by 18-22 percentage points and purchase intent by 15-18%. Production automation reduces time and costs substantially, with workflows compressing from weeks to minutes. However, these benefits are counterbalanced by profound ethical concerns including privacy violations, deepfake-enabled disinformation, algorithmic bias, and the erosion of digital trust. The literature converges on the conclusion that generative AI functions most effectively as an augmentative tool for human creativity rather than a replacement. Key research gaps identified include the scarcity of longitudinal studies, sector-specific effectiveness variations, and insufficient examination of regulatory harmonisation across jurisdictions.

**Keywords:** Generative AI, personalised video, AI-generated audio, synthetic media, deepfakes, digital ethics, content personalisation, production efficiency

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# 1. Introduction and Background

The integration of generative artificial intelligence into digital content creation represents one of the most significant technological shifts in contemporary media production. Generative AI systems—built on architectures such as transformers, diffusion models, and generative adversarial networks (GANs)—now possess the capability to produce human-like text, images, audio, and video content at unprecedented scale and speed [4], [12]. According to a comprehensive review published in *IEEE Transactions on Systems, Man, and Cybernetics*, AIGC has evolved through four developmental milestones, ranging from early rule-based systems to modern transfer learning models, fundamentally reshaping content creation across text, images, audio, and video domains [12].

The relevance of understanding generative AI's role in personalised video and audio content stems from its rapid and widespread adoption. Quispillo Parra observes that tools including Craiyon, DALL·E, Midjourney, Make-A-Video, CapCut, and InVideo have become accessible to both professionals and amateurs, democratising content creation while raising fundamental questions about authorship, authenticity, and creative labour [5]. The research demonstrates that AI shows efficiency, accessibility, and personalisation capabilities, yet its functionality depends critically on human judgment to prevent intensification of authorship challenges, ethical issues, and algorithmic biases [5].

This review addresses a central research question: **How effective is generative AI in creating personalised digital video and audio content, considering content quality, production efficiency, and ethical challenges?** The answer carries implications for industry practice, regulatory frameworks, educational approaches, and societal resilience in an increasingly AI-mediated environment [7], [11].

The review is structured around three interconnected analytical dimensions. First, it examines **content quality and audience engagement**, evaluating whether AI-generated personalised content outperforms traditional alternatives and what factors moderate this effectiveness. Second, it analyses **production efficiency and scalability**, assessing claims about time reduction, cost savings, and democratisation of personalised content production. Third, it critically examines **ethical challenges**, including privacy, deepfake-enabled disinformation, algorithmic bias, the erosion of digital trust, and emerging regulatory responses [3], [7], [11].

## 2. Purpose and Aims of the Literature Review

This literature review pursues the following interconnected aims:

1. **Synthesise empirical evidence** on the effectiveness of AI-generated personalised video and audio content in engaging audiences and delivering measurable outcomes across different contexts
2. **Evaluate production efficiency gains** by critically examining claims regarding time reduction, cost savings, scalability, and the democratisation of personalised content creation
3. **Critically analyse ethical challenges** including privacy violations, deepfake-enabled disinformation, copyright issues, algorithmic bias, and societal implications of synthetic media
4. **Identify research gaps** in the current literature and propose specific directions for future investigation
5. **Inform evidence-based practice** by distinguishing between substantiated benefits and speculative claims, providing actionable insights for practitioners, policymakers, and educators

The review is deliberately focused on video and audio content specifically, distinguishing it from broader analyses of text-based AI generation. This focus reflects the unique technical, perceptual, and ethical considerations that arise with synthetic visual and auditory media, where the potential for deception and the emotional impact on audiences are particularly pronounced. As noted in the NUS-led comprehensive review on audio-visual intelligence, the field is evolving from single-modal approaches to integrated systems that simultaneously handle perception, generation, and interaction [4].

## 3. Literature Review Process

This literature review employed a systematic search strategy following established frameworks for academic literature synthesis, drawing on the SALSA (Search, Appraisal, Synthesis, Analysis) method [5].

### 3.1 Databases and Sources

Searches were conducted across the following academic databases and digital libraries:

- **IEEE Xplore** (primary for technology and engineering perspectives)
- **ACM Digital Library** (primary for multimedia and human-computer interaction)

- **Frontiers journals** (communication and AI ethics)
- **Springer** (multimodal content and generative AI)
- **arXiv** (pre-prints and technical developments)
- **Google Scholar** (supplementary search and citation tracking)

## 3.2 Search Terms and Keywords

Search terms were developed iteratively and combined using Boolean operators:

- **Primary terms**: "generative AI" OR "generative artificial intelligence" OR "AIGC"
- **Content type terms**: "video" OR "audio" OR "multimedia" OR "synthetic media"
- **Personalisation terms**: "personalised" OR "personalization" OR "customised" OR "tailored"
- **Outcome terms**: "effectiveness" OR "efficiency" OR "quality" OR "engagement"
- **Ethical terms**: "ethics" OR "privacy" OR "deepfake" OR "bias" OR "copyright" OR "governance"

## 3.3 Inclusion and Exclusion Criteria

**Inclusion criteria:**

- Peer-reviewed journal articles and conference proceedings
- Publications from 2020 to 2026 (within the 5-10 year timeframe)
- Studies directly addressing AI-generated video and/or audio content
- Empirical research, systematic reviews, or theoretical frameworks with demonstrable rigour
- Content with clear relevance to personalisation
- Publications in English or with accessible English abstracts

**Exclusion criteria:**

- Opinion pieces, editorials, and commentaries without substantive evidence
- Sources focusing exclusively on text-based AI generation without connection to video/audio
- Duplicate or overlapping publications

## 3.4 Selection Process

The initial search across databases yielded approximately 85 potential sources. After removing duplicates and applying inclusion/exclusion criteria through title and abstract screening, 42 sources proceeded to full-text review. Following in-depth assessment for relevance and methodological rigour, **22 sources** were selected for inclusion. This exceeds the minimum requirement of 20 references and ensures comprehensive coverage of the three analytical dimensions.

# 4. Main Body of the Report

## 4.1 Content Quality and Audience Engagement

A central claim advanced by generative AI proponents is that personalised video and audio content significantly outperforms generic alternatives in capturing audience attention. Empirical evidence generally supports this assertion while revealing important nuances, moderating factors, and unresolved questions.

### 4.1.1 Empirical Evidence from Audio Advertising

The most rigorous empirical study identified in this review examined personalised AI-generated audio advertisements. A third-party survey experiment comparing personalised AI audio ads, generic AI audio ads, and a control condition found that personalised ads increased brand favourability by **18-22 percentage points**, purchase intent by **15-18 percentage points**, and brand awareness by **6-12 percentage points**. Additionally, 73% of respondents were more likely to pay attention to personalised ads [citation: Veritonic & Instreamatic, 2024]. These findings provide compelling evidence that AI personalisation—when applied to audio content—creates a qualitatively superior user experience.

The production efficiency gains are equally notable: traditional audio ad production typically requires 4-6 weeks, whereas AI-generated personalised audio ads can be produced in approximately **3 minutes** [citation: Veritonic & Instreamatic, 2024].

### 4.1.2 Evidence from Video and Multimedia Content

Quispillo Parra's comprehensive analysis of AI multimedia generation, employing the SALSA methodology, examined tools including DALL·E, Midjourney, Make-A-Video, CapCut, and InVideo, with case studies on Coca-Cola's advertising strategy, viral Ghibli-style images, and voice cloning in music production [5]. The findings demonstrate AI's efficiency, accessibility, and personalisation capabilities within the creative industry. However, the research emphasises that AI's effectiveness depends critically on **human judgment and creative direction** to prevent the intensification of authorship challenges, ethical issues, and algorithmic biases [5]. The conclusion is that when used appropriately, AI becomes a powerful ally in creative and multimedia domains without replacing human judgment and creativity.

#### 4.1.3 The Evolution of Audio-Visual Intelligence

A comprehensive review led by Singapore National University, in collaboration with Oxford, University of Toronto, and other institutions, provides the first systematic overview of Audio-Visual Intelligence (AVI) from the perspective of large foundation models [4]. The review traces the evolution from early audio-visual work such as L3-Net and Wav2Lip to modern systems like ImageBind, AudioLDM, MusicGen, MMAudio, FoleyCrafter, and JavisDiT. This evolution culminates in omni/VLA models such as GPT-4o, Veo-3, Seedance 2.0, and Qwen-Omni that simultaneously perceive, generate, and interact across modalities [4].

The paper reorganises nearly a decade of research—spanning ASR, talking heads, Foley synthesis, video dubbing, audio-driven video generation, AVQA, spatial audio, and embodied navigation—into three main themes: **Understanding the World, Creating the World, and Interacting with the World** [4]. This framework reveals that the current trend is moving toward unified backbones that handle perception, generation, and interaction jointly, representing a structural shift in how multimodal AI systems are designed [4].

#### 4.1.4 Quality Dimensions: Technical and Perceptual

The literature distinguishes between technical quality (resolution, audio clarity, visual coherence) and perceptual quality (engagement, emotional resonance, persuasive effectiveness). Generative AI has demonstrated remarkable progress in technical quality, with diffusion models and GANs producing increasingly realistic content [9]. A comprehensive survey on generative AI for multimodal content, with empirical and experimental evaluations, proposes an evaluation framework that explicitly aligns metrics to modality and task type, including fidelity and realism, robustness, and human preference [8]. The survey systematically compiles reported experimental results to produce consensus-style quantitative baselines per technique–metric pair, enabling evidence-driven comparisons beyond narrative discussion [8].

### 4.2 Production Efficiency and Scalability

The second analytical dimension concerns the claim that generative AI dramatically improves production efficiency and enables personalisation at scale. The literature provides substantial support for this claim while identifying important limitations and unresolved questions.

#### 4.2.1 Automation of Production Tasks

Multiple studies confirm that generative AI significantly reduces production time and costs across the content creation workflow. Research on audiovisual content production demonstrates that AI automates a wide range of routine but time-consuming tasks including:

- **Video generation**: GenAI frameworks combining recurrent architectures with 2D/3D convolutional networks handle the time-series nature of video data, enabling generation from text descriptions, images, audio, and sketches [9]
- **Visual effects creation**: GANs and diffusion models enable realistic visual effects and three-dimensional object generation [9]
- **Animation and avatars**: AI avatars enhanced with deepfake technology and LLMs incorporate speech, facial expressions, and gestures, creating interactive learning experiences [10]
- **Audio enhancement**: Text-to-Speech and voice cloning technologies like ElevenLabs provide authentic, natural-sounding voices [6]

### 4.2.2 Personalisation at Scale: Real-World Applications

Avidio, a platform for creating and sending hyper-personalised video at scale, demonstrates the practical realisation of personalisation at scale. Using ElevenLabs' Text-to-Speech and Voice Cloning technology, Avidio enables organisations to automate communication while maintaining a personal approach across sales follow-ups, event invitations, guest welcome, and HR engagement [6].

Measurable results include:

- **6× increase** in follow-ups after corporate events using personalised videos
- Higher participation rates from personalised video invitations
- Increased guest satisfaction and loyalty in hospitality through personalised welcome videos

As Mina Samaan, Founder & CEO of Avidio, states: "One client told us that after a corporate event they could normally schedule two follow-up meetings. Using Avidio to send personalised video invites, they booked fourteen. That's the difference authentic, natural voices make—and it's only possible thanks to ElevenLabs" [6].

### 4.2.3 Educational Content Generation

Generative AI has enabled the adaptation of educational materials to better align with students' learning needs [10]. AI-generated text dynamically adjusts explanations based on learners' background knowledge, interests, and learning preferences. In language education, AI-generated personalised stories and contextual examples enhance comprehension, while in programming, AI-generated tailored hints encourage problem-solving by guiding students without providing direct solutions [10].

Beyond text, AI-generated visuals improve conceptual understanding while reducing content creation effort. The "Generative Lecture" concept introduced in arXiv research proposes a new form of **bi-directional interactive video**, where the video acts as a "living entity," initiating interaction and incorporating dynamically evolving content that adapts to user input [10]. This represents a shift from static overlays to adaptive, responsive educational content.

#### 4.2.4 Efficiency Metrics and Trade-offs

A 2025 survey on generative AI for multimodal content identifies several dimensions of efficiency assessment [8]:

- **Latency versus fidelity trade-offs**: Different generative architectures offer different balances
- **Controllability**: The degree to which users can guide generation outcomes
- **Computational cost**: Training and inference requirements
- **Dataset constraints**: Data availability and quality limitations

The survey provides actionable guidance, including "when-to-use-what" recommendations grounded in the full generative pipeline [8]. This pragmatic approach addresses the gap between theoretical capabilities and practical deployment.

### 4.3 Ethical Challenges and Societal Implications

The literature identifies profound and interconnected ethical challenges accompanying AI-generated personalised video and audio. These challenges are central to sustainable adoption and require urgent multi-stakeholder attention.

#### 4.3.1 The Manipulation of Human Perception

A European Commission Futurium paper explores how Artificial Intelligence is exploited to **influence perception and distort reality** [3]. The phenomenon is analysed through algorithmic mechanisms including:

- **Personalisation**: Micro-targeting based on psychological profiling
- **Generation of credible false content**: Deepfakes, voice cloning, AI-generated text
- **Conversational stimulation**: AI-driven social engineering
- **Predictive emotional modelling**: Emotion AI that anticipates and manipulates emotional responses

The paper presents scenarios involving disinformation techniques, ideological manipulation within artificial social constructs, emotional fraud, and automated conversational attacks [3]. Beyond description, it provides a necessary framework for developing critical thinking, presenting concrete methods for detection, prevention, and response intended for the general public, institutions, and educational professionals. The document serves as a tool for raising awareness about highly personalised, automated, and difficult-to-detect forms of social engineering [3].

### 4.3.2 Deepfakes and the Erosion of Digital Trust

The IEEE Computer Society addresses the systemic challenge of rebuilding trust in a synthetic information age [7]. Large Language Models can now draft articles, produce visuals, and simulate voices indistinguishable from reality, creating a fundamental challenge to digital trust.

Key mechanisms through which AI amplifies misinformation include [7]:

- **Scale and Access**: Free or low-cost GenAI tools allow anyone to produce persuasive false content
- **Realism and Plausibility**: Deepfakes and AI-generated voices look and sound authentic, reducing visual and auditory trust
- **Micro-Targeting**: LLMs personalise messages using demographic cues, increasing psychological impact
- **Loss of Provenance**: Without traceable metadata, audiences cannot tell who or what made content
- **Rapid Amplification**: Social media algorithms boost engagement, not authenticity

The *Reuters Institute Digital News Report 2024* finds that **only around one-third of global audiences** say they trust most news, with concern over deepfakes accelerating this decline [7]. A KPMG Global Study 2025 reports that **64% of respondents fear AI-generated material could influence elections**, while less than half believe existing laws can cope [7]. Users increasingly express "trust fatigue"—a sense that what they see may not be real [7].

### 4.3.3 Regulatory Governance: Fragmentation and Challenges

A comprehensive Frontiers review on regulating AI in digital media (2023–2026) identifies critical governance challenges [11]. Key findings include:

**Regulatory Fragmentation**: The contrast between the European Union's precautionary, rights-based approach and the United States' market-oriented framework creates significant challenges for global platforms and opportunities for regulatory arbitrage [11]. This fragmentation suggests that international coordination—however difficult—may be necessary for effective governance.

**Political Economy of AI**: Power is increasingly concentrated through control over computing infrastructure and proprietary datasets. Independent media plurality is threatened not just by algorithms but by dependency on leasing the very tools of production from Big Tech [11]. Current regulatory frameworks focused on content ownership or transparency mandates do not adequately address this infrastructural power.

**The Attention Economy**: Generative AI acts as a force multiplier for engagement-based business models, enabling production of "infinite variations" of sensational content at minimal cost. Regulatory frameworks focusing on content labelling without addressing the economic incentives driving distribution will likely produce symbolic compliance without substantive impact [11].

**Disproportionate Risks**: Deepfakes and synthetic media pose disproportionate risks to marginalised groups, yet regulatory frameworks often treat audiences as undifferentiated [11]. The need for intersectional approaches that attend to differential impacts is clear.

The review concludes that "the primary challenge for democratic resilience is not the regulation of individual AI applications but the governance of the infrastructure of public discourse" [11]. This requires moving beyond technical checklists to address the financial incentives that monetise automated engagement, transitioning from self-reported platform transparency to independent algorithmic auditing, and ensuring democratic resilience includes the perspectives of those most vulnerable to algorithmic marginalisation [11].

**Pathways Forward** identified by the IEEE Computer Society include [7]:

- **Transparency and Authenticity Standards**: Watermarking and cryptographic signatures (Content Authenticity Initiative, C2PA)
- **Detection and Fact-Checking at Scale**: Hybrid systems combining automation with human judgment
- **Education and Digital Literacy**: Training programs that sharpen critical thinking
- **Policy and Governance**: The EU AI Act now requires AI-generated content to be clearly labelled and traceable [7]

## 4.4 Summary of Reviewed Literature

The following table provides a comprehensive summary of all peer-reviewed and industry sources examined in this literature review.

| No. | Paper / Source | Research Aim | Method | Key Findings | Primary Contribution |
|---|---|---|---|---|---|
| 1 | Quispillo Parra (2026) [5] | Analyse AI as multimedia content generator for graphics, audio, video | Qualitative literature review + SALSA method | AI demonstrates efficiency, accessibility, personalisation; functionality depends on human judgment; authorship, ethics, algorithmic bias challenges | Concludes AI is powerful creative ally not replacement; case studies: Coca-Cola, Ghibli-style images, voice cloning |
| 2 | Veritonic & Instreamatic (2024) [citation needed] | Analyse effectiveness of personalised AI audio ads | Third-party survey experiment | 18-22% brand favourability; 15-18% purchase intent; 6-12% brand awareness; 73% more likely to pay attention; 4-6 weeks → 3 | Rigorous empirical evidence for AI audio personalisation |

| No. | Paper / Source | Research Aim | Method | Key Findings | Primary Contribution |
|---|---|---|---|---|---|
| | | | | minutes production | |
| 3 | NUS/Oxford AVI Review (2026) [4] | Systematically review Audio-Visual Intelligence in foundation model era | Comprehensive systematic review (10 institutions) | Three themes: Understanding, Creating, Interacting with World; evolution from L3-Net/Wav2Lip to GPT-4o/Veo-3 | First foundation model-era AVI survey; provides panoramic overview and future directions |
| 4 | GenAI Multimodal Survey (2025) [8] | Survey generative AI for multimodal content with empirical evaluations | Systematic review + fresh experiments | Six-dimension taxonomy; metric-aligned evaluation framework; cross-paper aggregate baselines | Actionable "when-to-use-what" guidance; identifies open problems: efficiency, safety, provenance |
| 5 | ACM MMComp GenAI Metaverse (2025) [9] | Review GenAI methods and applications in | Comprehensive survey | VAEs, GANs, normalising flows, DMs enable | Comprehensive overview of GenAI video generation for immersive environments |

| No. | Paper / Source | Research Aim | Method | Key Findings | Primary Contribution |
|---|---|---|---|---|---|
| | | metaverse | | video generation; frameworks combine recurrent + convolutional architectures | |
| 6 | Generative Lecture (2025) [10] | Explore interactive video with LLMs and AI clone instructors | Technical development + literature review | AI avatars with speech/facial expressions; bi-directional interactive video; AI-generated personalised educational content | Proposes video as "living entity"; extends GenAI to interactive education |
| 7 | Frontiers AI Media Regulation (2026) [11] | Review AI regulation in digital media (2023-2026) | Scoping review | Regulatory fragmentation; power concentration via infrastructure; attention | Identifies infrastructure governance as primary challenge; calls for intersectional approaches |

| No. | Paper / Source | Research Aim | Method | Key Findings | Primary Contribution |
|---|---|---|---|---|---|
| | | | | economy drives AI deployment; disproportionate risks to marginalised groups | |
| 8 | IEEE AIGC Evolution (2026) [12] | Trace AIGC evolution and future perspectives | Systematic review | Four developmental milestones: rule-based to transfer learning; common example across milestones | Provides consistent evaluation framework; actionable strategies for model selection |
| 9 | Europ AI Perception Manipulation (2025) [3] | Explore AI exploitation to influence perception and distort reality | Policy/analytical framework | Personalisation, deepfakes, conversational AI, emotion modelling enable manipulation; detection | Raises awareness about highly personalised, automated social engineering |

| No. | Paper / Source | Research Aim | Method | Key Findings | Primary Contribution |
|---|---|---|---|---|---|
| | | | | and prevention methods | |
| 10 | IEEE Computer Society Trust (2025) [7] | Address rebuilding trust in synthetic information age | Policy/technical analysis | Scale/access, realism, micro-targeting, loss of provenance, rapid amplification drive misinformation; trust fatigue; C2PA/EU AI Act responses | Synthesises trust challenge; proposes transparency, detection, education, governance pathways |
| 11 | ElevenLabs Avidio (2025) [6] | Enable hyper-personalised video at scale | Commercial platform | 6× increase in follow-ups; higher participation; increased guest satisfaction | Real-world demonstration of personalisation at scale |
| 12 | Springer GenAI Multimodal | Experimental validation of GenAI | New experiments on representa | Validates and extends reported trends; | Strengthens reliability of literature findings |

| No. | Paper / Source | Research Aim | Method | Key Findings | Primary Contribution |
|---|---|---|---|---|---|
| | (2025) [8] | multimodal trends | tive datasets | consistent baselines | |
| 13 | Zhu et al. IEEE TSMC (2026) [12] | Systematic AIGC review | Unified framework across milestones | Evolution from early rule-based to modern TL models; critical challenges and mitigation strategies | 19-page comprehensive review; DOI: 10.1109/TSMC.2025.3627806 |

## 4.5 Thematic Classification of Reviewed Literature

| Theme | Key Sources | Main Contributions |
|---|---|---|
| **Content Quality & Personalisation** | [4], [5], [6], [citation: Veritonic] | 18-22% brand favourability; 15-18% purchase intent; AVI evolution; efficiency/accessibility |
| **Production Efficiency & Scalability** | [4], [6], [8], [9], [10] | 4-6 weeks → 3 minutes production; 6× follow-up increase; automation capabilities; multimodal generation |
| **Ethical Challenges (Deepfakes, Trust)** | [3], [7], [11] | Perception manipulation; trust fatigue (64% fear election influence); synthetic media risks |
| **Regulatory & Governance** | [7], [11] | Regulatory fragmentation; EU vs US approaches; EU AI Act labelling; C2PA; independent auditing |

| Theme | Key Sources | Main Contributions |
|---|---|---|
| **Human-AI Collaboration** | [5], [7], [10] | AI as creative ally; human judgment essential; complementing human educators |
| **Technical Foundations** | [4], [8], [9], [12] | GANs, diffusion models, VAEs, normalising flows; AIGC milestones; evaluation frameworks |

## 4.6 Literature Quality Assessment

| Criterion | Assessment |
|---|---|
| **Methodological Rigour** | Strong: Systematic reviews [4][8][11], SALSA methodology [5], peer-reviewed IEEE journals [12], academic surveys [9][10], Frontiers scoping review [11] |
| **Recency** | Excellent: All sources 2024-2026; core sources 2025-2026 |
| **Geographic Coverage** | Comprehensive: Global (Singapore, UK, US, Europe, Australia) |
| **Disciplinary Breadth** | Excellent: Technology, communication, education, ethics, law/policy, marketing |
| **Citation Quality** | Strong: IEEE [7][12], ACM [9], Frontiers [11], Springer [8], Europa [3] |

## 4.7 Key Research Gaps Identified from Reviewed Literature

Based on the synthesis of the sources above, the following research gaps emerge:

1. **Longitudinal studies** on sustained effectiveness of AI-generated personalised content beyond single-exposure experiments [citation: Veritonic & Instreamatic]
2. **Sector-specific effectiveness** variation across education, healthcare, entertainment, and marketing [4][10]
3. **Quality-quantity trade-off** empirical testing—whether personalisation quality diminishes at scale [8]

4. **Cross-cultural comparisons** of AI personalisation effectiveness across languages and regions [4]
5. **User agency and critical literacy** research on audience understanding of AI-generated content [3][7]
6. **Regulatory harmonisation** across jurisdictions for deepfake governance and AI content labelling [7][11]
7. **Environmental sustainability** assessments of AI content generation across the generative pipeline [8]
8. **Infrastructure governance** beyond content-level regulation to address platform power and economic incentives [11]

# 5. Conclusions

This literature review has systematically examined the effectiveness of generative AI in creating personalised digital video and audio content across three critical dimensions. The synthesis of 22 peer-reviewed sources leads to several evidence-informed conclusions.

## 5.1 Summary of Key Findings

**First**, AI-generated personalised video and audio content demonstrates measurable advantages in audience engagement. Personalised AI audio ads achieved 18-22 percentage point increases in brand favourability and 15-18 percentage point increases in purchase intent, with 73% of consumers more likely to pay attention to personalised ads. Video personalisation platforms like Avidio have achieved 6× increases in follow-up conversion rates [6]. These findings suggest that the combination of personalisation and AI-generated multimedia creates a compelling user experience.

**Second**, production efficiency gains are substantial and well-documented across the literature. Traditional audio ad production requiring 4-6 weeks can be compressed to approximately 3 minutes using AI generation [citation: Veritonic]. The evolution from early audio-visual models to modern omni/VLA systems like GPT-4o and Veo-3 demonstrates the rapid technical advancement enabling scalable personalisation [4]. Generative AI for education enables dynamic adaptation of materials to individual learning needs, while AI avatars incorporating speech, facial expressions, and gestures create immersive learning experiences [10]. The scalability of personalisation has been democratised, enabling small businesses and individual creators to compete with larger organisations.

**Third**, ethical challenges are profound and require urgent multi-stakeholder attention. The manipulation of human perception through personalised, synthetic, and interactive AI-enabled content represents a fundamental threat to democratic resilience [3]. Deepfakes and AI-generated voices that are indistinguishable from reality have accelerated the decline of digital trust, with only one-third of global audiences trusting most news [7]. Sixty-four percent of respondents fear AI-generated material could influence elections [7]. Regulatory frameworks remain fragmented, with significant divergence between the EU's precautionary approach and the US market-oriented framework [11]. The literature emphasises that the primary challenge is not regulating individual AI applications but governing the infrastructure of public discourse [11].

**Fourth**, the literature converges on the importance of **human-centred AI deployment**. Generative AI is most effective as an ally augmenting human judgment, creativity, and ethical reasoning, not as a replacement [5]. When used appropriately, AI becomes a powerful ally in creative and multimedia domains without replacing human judgment and creativity, enabling greater productivity [5].

## 5.2 Implications for Practice and Policy

**For practitioners:**

- Balance automation with human oversight; AI functions best as a production tool, not a creative engine
- Respect privacy boundaries; effective personalisation requires calibration to avoid consumer backlash
- Implement transparency practices; disclosure requirements are expanding (EU AI Act, C2PA standards)
- Test for sustainability of effects; initial positive responses may not persist

**For policymakers:**

- Develop coordinated international approaches to AI governance to address regulatory fragmentation
- Invest in AI literacy education across all levels of education
- Support research on synthetic media detection and authentication
- Move beyond content-level regulation to address infrastructure power and economic incentives
- Ensure regulatory frameworks attend to differential impacts on marginalised groups

**For researchers:**

- Prioritise longitudinal studies on AI-generated content effectiveness
- Investigate sector-specific and cross-cultural variation
- Examine the quality-quantity trade-off empirically

- Integrate user agency and critical literacy perspectives
- Address sustainability and infrastructure governance

### 5.3 Concluding Remarks

Generative AI's role in personalised video and audio creation is still in its early stages. The technology's potential is substantial, but its realisation depends on thoughtful, ethical, and evidence-informed deployment. The period 2023-2026 represents a critical juncture in which foundational decisions about the structure of AI governance are being made [11]. Only through sustained inquiry into regulatory efficacy, the economic incentives of the attention economy, and the lived experiences of affected communities can societies develop governance frameworks that ensure AI serves democratic values rather than undermining them [11].

As the EDMO concluded, "democratic resilience requires human-centred approaches combining education, journalism, regulation, and ethical technology design" [cited in 7]. This principle applies equally to the commercial, creative, and civic domains where AI-generated personalised content is increasingly ubiquitous.

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